Training-Free Token Reduction on MotionBench

Results

VLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4{,}018$ the binomial 95% confidence half-width is ± 1.54 points; * marks a difference significant by McNemar's test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

Contents

1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 `NA` items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 6 (§4).

2 Master table

Table 1 and Table 2: every method at its standardized 15% or published default, one table per backbone. § AIM's released code ships two active merge steps, so its LLaVA-OV cell runs at 25%. MDP3 and VideoITG select frames, not tokens, so they have no retention percentage. ‡ DyCoke and STTM have no retention knob at all — each runs its one fixed published recipe (measured net 49% and 9%, Table 6), not a 15% target; kept in these tables because they are standard-backbone, gate-clean runs, unlike the excluded non-standard-backbone methods (§6). Full retention sweeps are in §2.1.

Table 1: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. baseline. * = significant. § AIM runs at 25% retention (two active merge steps ship in the released code). † PruneVID-OV runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%. ‡ no retention knob; one fixed recipe (Table 6). ¶ VisionZip is the contextual half only — LLaVA-OV's SigLIP vision tower has no CLS token, so the dominant half is not implementable (same limitation as the Qwen3-VL cell, Table 2); this LLaVA-OV port is new in this revision.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*52.66*	—	*40.5*	*45.2*	*55.5*	*57.0*	*71.2*	*23.8*
DyCoke‡	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.2
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.2
VisionZip (ctx.)¶	51.79	–0.87	40.1	46.2	53.7	56.2	69.4	23.2
STTM‡	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.7
PruneVID-OV*†	38.20	–14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	–15.88	32.8	31.9	34.1	35.7	53.8	25.2

Table 2: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only — Qwen3-VL's vision tower has no CLS token, so the dominant half is not implementable; not the published method. FlashVID and STTM run the authors' own code at published settings (§4). † PruneVID runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%. ‡ no retention knob; one fixed recipe (Table 6).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*62.52*	—	*46.1*	*63.1*	*65.0*	*67.3*	*79.0*	*33.8*
PruneVID†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke‡	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM*‡	61.60	–0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom*	60.33	–2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	–2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID*	59.36	–3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV*	59.01	–3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	–3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG*	56.35	–6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	–6.55	45.9	56.6	59.5	58.3	72.2	27.0

2.1 Retention sweep

Three methods (FastV, FlashVID, HoliTom) have a retention knob the authors validate at more than one point; each is run at every point they publish, not just 15%. Table 3 and Table 4 are that data, one table per backbone; baseline is in Table 1/ 2 above, not repeated here.

Table 3: LLaVA-OV-7B retention sweep (baseline 52.66%). * = significant. FastV's range is its authors' full published boundary, re-run with the `PrunableDynamicCache` fix (§4) — every point reproduces its pre-fix value almost exactly. FlashVID and HoliTom publish 10/15/20/25% on every backbone they support; VisionZip (contextual-only, §2) was swept over the same range for this report.

Method	10%	15%	20%	25%	50%	75%
FastV	36.06*	36.78*	36.93*	36.73*	36.31*	35.69*
FlashVID	52.51	53.31	53.71*	53.36	—	—
HoliTom	52.76	53.14	53.06	53.41	—	—
VisionZip (ctx.)	51.69	51.79	51.92	52.54	—	—

Table 4: Qwen3-VL-8B retention sweep (baseline 62.52%). * = significant. All four FlashVID cells now use the authors'-code implementation (§4), the same one that produced the 15% cell elsewhere in this report — monotonic in retention. FlashVID and HoliTom are not published beyond 25% on this backbone.

Method	10%	15%	20%	25%	50%	75%
FastV	56.92*	59.01*	59.83*	60.60*	62.32	62.64
FlashVID	58.04*	59.36*	60.05*	60.70*	—	—
HoliTom	60.05*	60.33*	60.38*	60.43*	—	—

FlashVID's 20% cell is a small local peak on LLaVA-OV (53.71% vs. 53.31%/53.36% at 15%/25%) but not on Qwen3-VL, where all four points are now monotonic (58.04/59.36/60.05/60.70% at 10/15/20/25%) since every Qwen3-VL FlashVID cell uses the same authors'-code implementation. The LLaVA-OV bump is gate-clean but does not persist outside 20%, so it does not change the picture that FlashVID is roughly flat on that backbone — on Qwen3-VL it instead responds cleanly and monotonically to retention. HoliTom's 20% shows no such bump on either backbone. VisionZip's new LLaVA-OV sweep (51.69–52.54%) is gentle and never significant at any point — unlike FastV and PruneVID-OV, it does not collapse on this backbone. FastV is the sharpest backbone contrast in this report: at 50%/75% keep it stays collapsed on LLaVA-OV (35.7–36.3%, Table 3) but fully recovers to baseline parity on Qwen3-VL (62.32%/62.64%, both not significant against the 62.52% baseline) — the same pruning rule that permanently breaks the model on one backbone is harmless by half retention on the other.

PruneVID publishes exactly one operating point (`cluster_ratio=0.5`, its row in Table 1/2); Table 5 is the additional sweep run for context, not an authors'-validated point.

Table 5: PruneVID beyond its one published point (not authors'-validated). Qwen3-VL is monotonic in retention; LLaVA-OV is not.

Backbone	10%	25%	50% (published)
LLaVA-OV-7B (OV port)	38.10*	38.38*	38.20*
Qwen3-VL-8B	60.23*	61.40*	62.17

Tripling FastV's budget from 10% to 25% moves it 3.68 points on Qwen3-VL, against no more than 0.85 points for any LLaVA-OV method over the same range — the two backbones respond to retention completely differently.

3 Figures

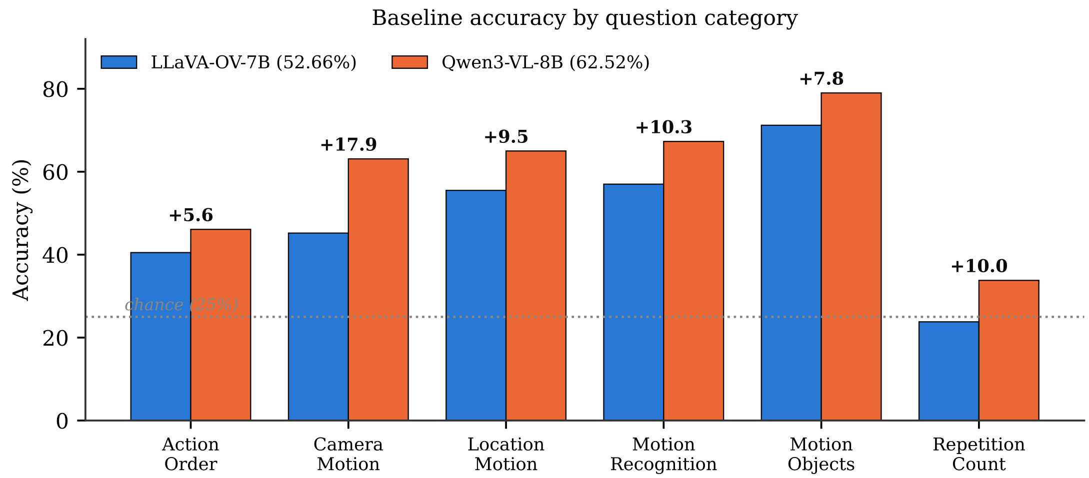

Figure 1: Baseline accuracy by category, both backbones (bar).

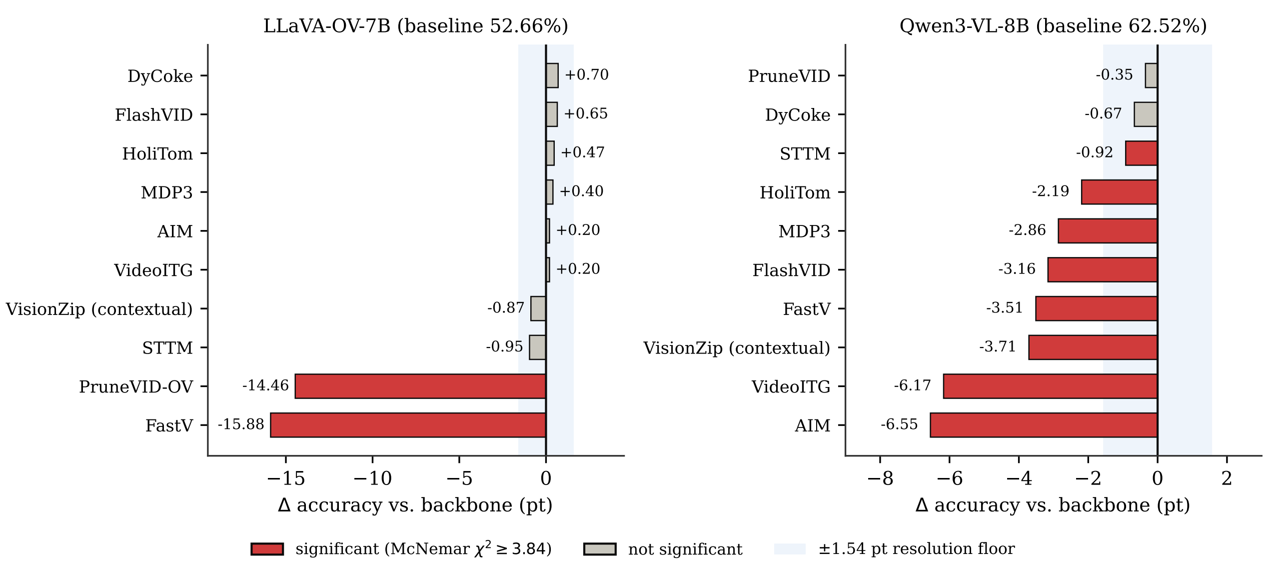

Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar's test; the shaded band is the ±1.54-point resolution floor.

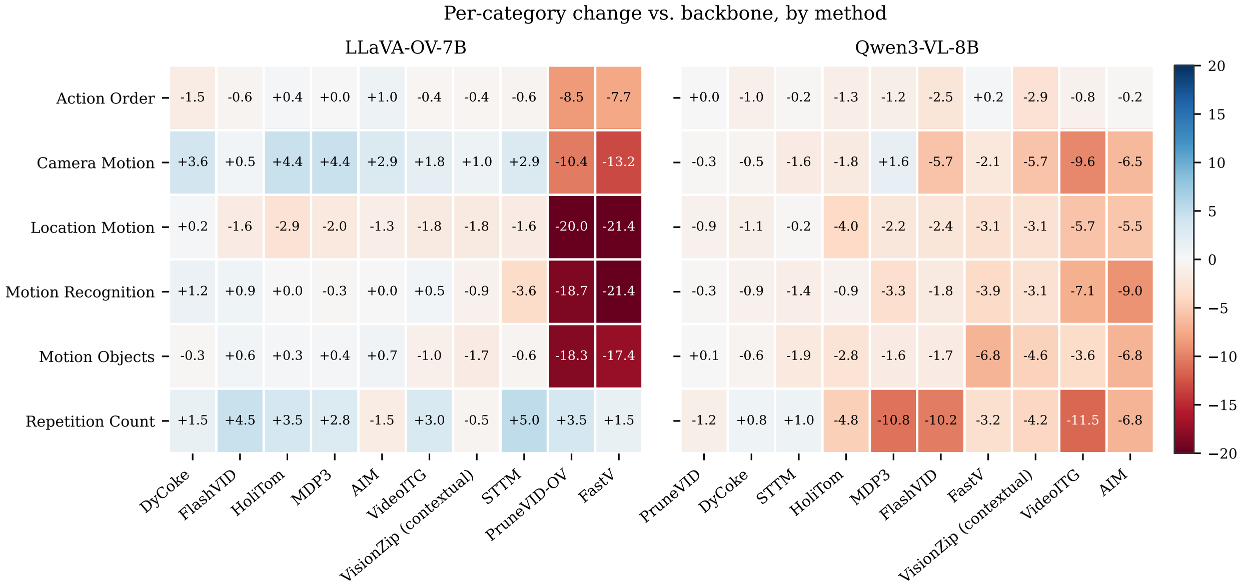

Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

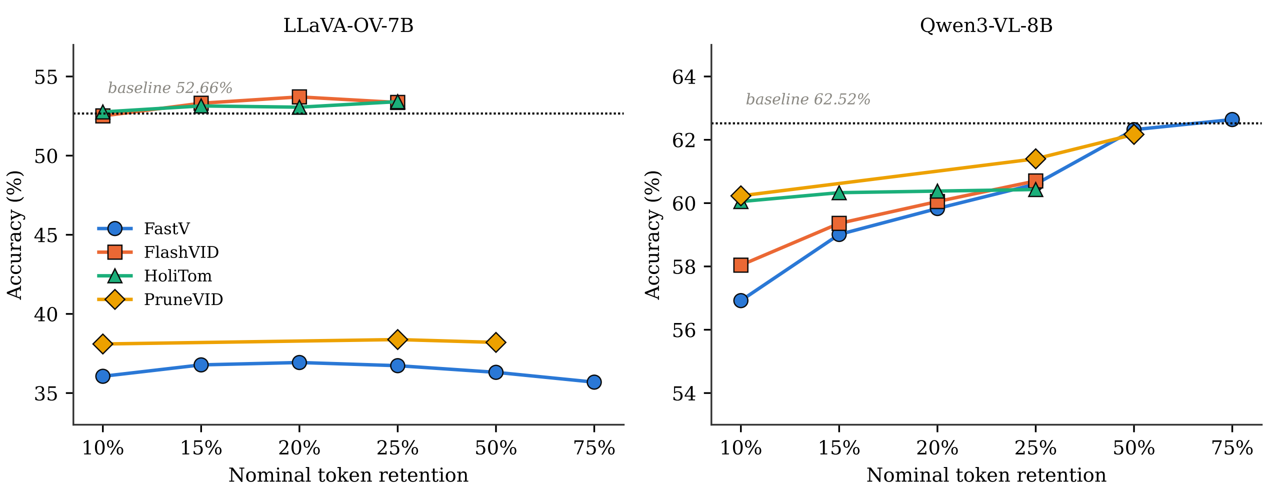

Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

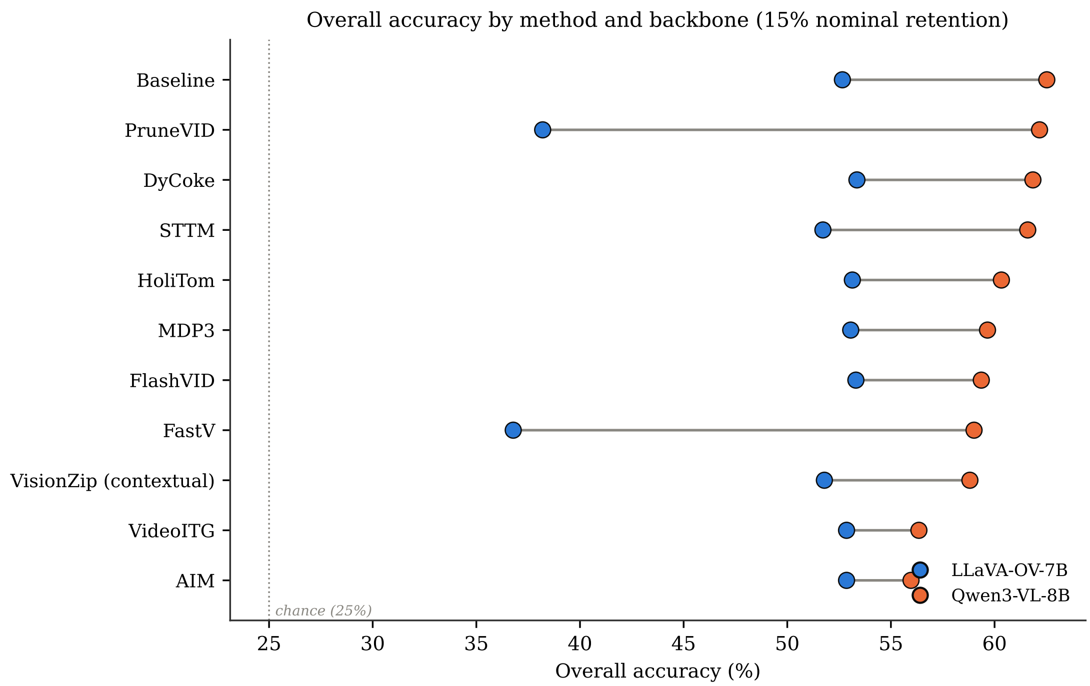

Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method's two backbone results.

4 Per-method settings

Table 6 is the mechanism and exact configuration behind every row above.

Table 6: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method's execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	`RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5`	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: `thresh=0.85 temporal=0.65 root=1`. Qwen3-VL: `thresh=0.85 temporal=0.65 root=2` — a published setting (`s3_sttm_pub_run`), replacing an earlier off-spec `root=0, temporal=--1.0` cell	9.0% (1,050 tok.) on LLaVA-OV; not re-measured at the corrected Qwen3-VL setting
FastV[‡]	in-LLM attention rerank, layer 2	`K=2, R=0.85` (keep 15%). Re-run with the `PrunableDynamicCache` fix; K=2/R=50% (36.31%) and K=2/R=75% (35.69%) were also run and land in the same collapse band (Table 3)	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge + inner-LLM compression, layer 28	authors' `flashvid()` package: `retention_ratio=0.15 alpha=0.7 temporal_threshold=0.8 token_selection_method=attn_div pruning_layer=28 llm_retention_ratio=0.1 expansion=1.25 min_segment_num=4`	15.0% (1,750 tok.) pre-LLM; the layer-28 stage compresses surviving visual tokens by a further 90% mid-network, not captured by this single figure
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	LLaVA-1.5-7B (native): `dominant=54 contextual=10` at 8 frames. LLaVA-OV & Qwen3-VL: contextual-only port, `retention_ratio=0.15` — neither vision tower has a CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	`l=3 p=0.7 k=0.7`; no single retention knob, net ∼49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	`cluster=0.50 seg=0.25`; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	`pool=32 select=8` — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo[†]	FINCH clustering + ToMe token merge	`temporal_aggregation=spatial_tome_finch_dynamic_all_frms`; own Vicuna-7B backbone	∼3,680 tok. from 100 input frames

[†] DyTo does not run on either standardized backbone; listed here for completeness (see Table 8). [‡] every prior LLaVA-OV FastV run silently discarded its own pruning (a `PrunableDynamicCache` bug, §7); re-run with the fix, every point reproduces its pre-fix value almost exactly.

5 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ* counts predictions unequal to the baseline's over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 7: Paired-prediction statistics against each backbone's baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
LLaVA-OV-7B					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
VisionZip (ctx.)	1,725	260	295	2.08	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,701	475	1,113	255.52	yes
Qwen3-VL-8B					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
Backbone vs. backbone					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	810	414	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV's 4,701 — yet loses 0.95 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

6 Methods on non-standard backbones

Table 8: Runs on backbones other than the two standardized ones. Not comparable to Tables 1–2; no Δ is given because no matched baseline exists for any row. PruneVID's released code targets PLLaVA only — this is the authors' real backbone and the one cell with an upstream reference to validate against, unlike the LLaVA-OV port (PruneVID-OV, Table 1).

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
PruneVID	PLLaVA-7B	44.13	35.6	33.0	41.0	47.0	63.3	26.5
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

7 Notable

The backbone swap moves accuracy 9.86 points — more than any token-reduction method moves it, either direction, either backbone, any retention level tested. Rankings do not transfer: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (-0.35); Figure 5 shows the crossing lines directly. FastV and PruneVID-OV collapse specifically on LLaVA-OV (37–38%, > 14 points below baseline, significant), while every other LLaVA-OV method is statistically indistinguishable from the backbone; FastV's own published range does not save it (35.7–36.3% at 75% and 50% keep, Table 3). Qwen3-VL inverts the pattern: every method loses, eight of ten losses are significant, but none approaches the LLaVA-OV collapse — the largest Qwen3-VL loss (AIM, -6.55) is smaller than the smallest LLaVA-OV collapse.

Only three methods qualify as having an authors'-validated retention sweep (§2.1); the other five with any retention-like parameter do not. **PruneVID** publishes exactly one ratio. **DyCoke** has no single retention knob, only one fixed three-parameter recipe. **AIM** has no knob at all on LLaVA-OV and an unpublished, self-ported one on Qwen3-VL. **STTM** is configured per-benchmark, not a sweepable percentage. **VisionZip**'s contextual-only port does have a sweepable ratio, but the authors never validate one — our 10–25% sweep (Table 3) is ours, not theirs.

A cache bug silently discarded FastV's and PruneVID's pruning on LLaVA-OV. Both methods prune visual tokens by setting a kept-index list (`kv_cache`) on `PrunableDynamicCache`, the shared cache class this project's LLaVA-OV ports use. Two independent bugs in that mechanism meant the pruning never reached the generated text. First, the class's `update()` gathered the kept indices only for the tensors *returned* to attention that call — the tensors it *stored* stayed at full length, so at the end of the first `generate()` step (where `past_key_values` arrives as `None`) the cache was serialized back out via `to_legacy_cache()` at full, unpruned length, and every following decode step reconstructed a fresh, fully unpruned cache from that export. Second, PruneVID's loader separately passed `dycoke=True` (copied from the DyCoke loading template and never removed); with it set, the shared decoder layer loop resets `kv_cache` to `None` on every layer before PruneVID's own prune layer, *on every decode step*, which both erased PruneVID's pruning again and triggered DyCoke's own unrelated pruning path against stale state, crashing a large fraction of samples once the first bug was fixed and pruning stopped being silently discarded. Fixed by making the prune physically permanent in cache storage (survives the `to_legacy_cache` round-trip) and by removing the stray `dycoke=True`. Re-running FastV's full published range after the fix reproduces every pre-fix accuracy point to within 0.03%: the LLaVA-OV collapse documented above is confirmed to be genuine model behavior, not a plumbing artifact — fixing the bug does not change the result. PruneVID's LLaVA-OV numbers in this report predate the fix and are being re-run; neither Qwen3-VL port is affected (§4 documents the Qwen3-VL implementations as always using an additive attention mask, never this cache class).